SUSHMA KARNATI

AI Developer

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AI Developer with hands-on experience in developing and deploying **AI, NLP, Computer Vision, and Generative AI** and **LLM-based AI agent** models at scale using **Python, Docker**, and cloud technologies (**GCP/AWS**). Skilled in building and optimizing **LLM-powered solutions**, developing intelligent systems, and leveraging AI agents for automation and problem-solving in real-world applications.

Technical Skills

Languages & Frameworks: Python, TensorFlow, LangChain, JavaScript, PHP
Web Technologies: Html, Css, React JS, Bootstrap, Django, Flask, REST API Development, API Design & Integration, WebSockets, FastAPI
DataBases & Cloud: SQL, Excel, MySQL, Google Cloud Platform (GCP), AWS, Database Management System, PostgreSQL, Cloud-Based Model Deployment, Debugging, Documentation
AI & ML: AI, Machine Learning, Deep Learning, LLMs, Generative AI, NLP, Computer Vision, LLM APIs (OpenAI, Gemini), RAG (Retrieval-Augmented Generation), Prompt Engineering & Optimization, AI Agent Design (LangChain, AutoGen), Structured Outputs & Tool Calling, Model Deployment, Model Evaluation & Optimization
Software Engineering: Algorithms, Data Structures, Performance Optimization, Full Stack Software Development, Logic Development, Code Review, Core Backend Programming
DevOps & Tools: Docker, GitLab, JSON, CI/CD Pipelines, Performance Optimization, Code Reviews
Soft Skills: Analytical Thinking, Problem Solving, Cross-Functional Collaboration, Written & Verbal Communications, Logical Reasoning, Business Support Mindset, Ownership and Accountability, Customer and Stakeholder Management, Adaptability in Startup Environments, Continuous Improvement and Process Optimization

Experience

Arna Intelligence **April 2026 – Present**
AI Native Full Stack & Middleware Engineer *Hyderabad,India*

- **Designed** and built an **end-to-end RAG-based curriculum** alignment platform using **Supabase Edge Functions**, **React/TypeScript**, and **Python ingestion pipeline**s that embed **NCERT** and **US Common Core standards** with **vector search** to generate **personalized transition readiness reports**.
- **Engineered LLM prompt engineering**, structured **JSON** output parsing, **hallucination detection**, and **post-processing validation pipelines** for a **Gemini-powered AI agent** that retrieves **curriculum gaps**, **injects source URLs**, and produces **actionable bridge timelines** for students transitioning between education systems.

Telic Info Services Private Limited **November 2024 – December 2025**
Artificial Intelligence Developer *Chennai,India*

- **Designed** and **developed Agentic AI applications** for business automation using **LLMs**, **RAG** (Retrieval-Augmented Generation), **prompt engineering**, **tool calling**, and **verification/validation** pipelines to automate document processing, email handling, and operational workflows.
- **Built end-to-end AI solutions** using **Python (FastAPI/Flask)** by integrating **REST APIs**, implementing **AI agent workflows** with structured outputs, and developing scalable backend services for enterprise **business use cases**.

Projects

Autonomous FinOps Multi-Agent (FastAPI + LangChain + Ollama + ChromaDB)

- Built a **local, privacy-preserving** multi-agent financial operations system that ingests transaction events via **FastAPI**, performs **behavioral anomaly detection**, retrieves internal **fraud/compliance policies using RAG** (ChromaDB + HuggingFace embeddings), and **outputs an auditable structured JSON** risk report.
- Designed specialized agents **(Investigation, Policy/RAG, Decision, Action, Explanation)** orchestrated as an end-to-end workflow, enabling **deterministic risk scoring (0–10)**, **reason codes**, and **recommended actions** (approve / verify / hold / manual review).

- Implemented **safe autonomous workflow simulation** for downstream operations (customer verification notification, temporary hold, case creation) with **strong observability** (agents executed, tool calls, risk breakdown) and **LLM-generated explainability** (analyst summary + customer-safe message) using **Ollama Llama 3** with **no external API keys**.

Betty Assistant, AI Executive Assistant (React + Flask + GPT-4)

- Developed a **chat-based executive assistant** that manages email, calendar, tasks, and travel planning from one conversational interface.
- Built intent detection and parameter extraction using **OpenAI GPT-4**, including support for context, relative dates, and clear confirmations for user actions.
- Integrated **Gmail and Google Calendar** plus **Outlook and Outlook Calendar** using **OAuth 2.0** and **MSAL**, with reliable token refresh and session-based credential handling.
- Implemented **15+ REST APIs** for email actions, scheduling, travel search, and task CRUD, using SQLite for lightweight task storage and audit-friendly tracking.
- Shipped a responsive **React** UI with **Material UI**, Markdown rendering, and smooth API-driven interactions for daily executive usage.

GLOBICULUM, AI-Powered Curriculum Transition Platform (React + Supabase + RAG + Gemini)

- **Built Globiculum**, an **AI-driven** academic transition platform that maps **global curricula (US Common Core, NCERT)** to **Indian boards**, generating **personalized gap analysis** and bridge plans in **under 10 minutes.**
- **Architected a RAG pipeline** using **Supabase pgvector + Gemini embeddings** to semantically compare curriculum nodes and identify **critical learning gaps** across **subjects** and **grade bands**.
- **Developed Supabase Edge Functions (Deno/TypeScript)** for **secure, rate-limited AI analysis** with **JWT auth, request validation, hallucination checks**, and **structured JSON report generation** via **Gemini.**
- **Created a React + V**ite frontend with **interactive assessment flows, real-time alignment dashboards, knowledge graph visualizations, PDF report export**, and **persistent report history.**
- **Designed** an adaptive percentile-based gap scoring system that **classifies gaps as Critical/Major/Moderate** and produces phase-wise bridge timelines with resource-linked study recommendations; built to scale to **ICSE, state boards,** and **US state-specific curricula**.

Video-to-SOP, AI Documentation Automation (Gemini + FFmpeg + OpenCV)

- **Built** a **cloud-native system** that converts **training videos into a formatted Word SOP** with screenshots and a table of contents, reducing documentation effort by **~97%** (hours to minutes).
- Implemented video segmentation using **FFmpeg**, screenshot extraction with **OpenCV**, and duplicate-frame removal via **perceptual hashing** to keep outputs clean and lightweight.
- Used **Google Gemini 2.0** for multi-stage video understanding and step generation, applying prompt patterns to produce clear, action-oriented instructions.
- Deployed with **Docker** on **Google Cloud Run** with autoscaling and secure file handling through **Cloud Storage**, including lifecycle controls to reduce storage overhead.
- Added reliability features like structured logs, validation, retries, and failure handling for stable production operation.

Certifications

- **Conversational AI -** Wisely Wise
  Completed training on AI, ML, and Alexa, including building a chatbot and voice assistant using AWS.
- **Data Analytics -** Accenture
  Gained hands-on experience in data cleaning, modeling, visualization, and client presentation.
- **Generative AI -** DeepLearning.AI
  Understand the impacts of Gen AI on business and society to develop effective AI strategies and approaches.

Education

Bachelor of Technology in Artificial Intelligence & Machine Learning — **Sep 2020 – Jun 2024**
Jawaharlal Nehru Technological University Hyderabad — *Hyderabad*

- **GitHub -** github.com/karnatisushma